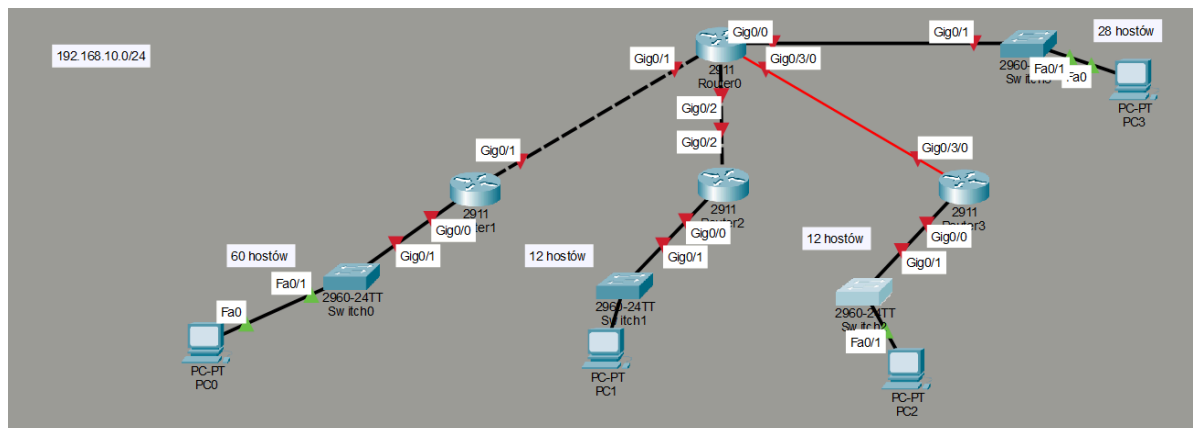


Cel:

Proszę podzielić sieć zgodnie z zapotrzebowaniem z pliku oraz dokonać konfiguracji urządzeń (hostname, secret, default gateway, OSPF)



Rozwiązanie:

Dzielenie sieci:

R1 G0/0:

/26 (255.255.255.192)

NET: 192.168.10.0

FH: 192.168.10.1

LH: 192.168.10.62

BR: 192.168.10.63

R0 g0/0:

/27 (255.255.255.224)

NET: 192.168.10.64

FH: 192.168.10.65

LH: 192.168.10.94

BR: 192.168.10.95

R2 g0/0:

/28 (255.255.255.240)

NET: 192.168.10.96

FH: 192.168.10.97

LH: 192.168.10.110

BR: 192.168.10.111

R3 g0/0:

/28 (255.255.255.240)

NET: 192.168.10.112

FH: 192.168.10.113

LH: 192.168.10.126
BR: 192.168.10.127

R0 g0/1
/30 (255.255.255.252)
NET: 192.168.10.128
FH: 192.168.10.129
LH: 192.168.10.130
BR: 192.168.10.131

R0 g0/2
/30 (255.255.255.252)
NET: 192.168.10.132
FH: 192.168.10.133
LH: 192.168.10.134
BR: 192.168.10.135

R0 g0/3/0
/30 (255.255.255.252)
NET: 192.168.10.136
FH: 192.168.10.137
LH: 192.168.10.138
BR: 192.168.10.139

Adresacja na sprzęcie:

R0:

```

Router>
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R0
R0(config)#enable secret 123
R0(config)#int g0/0
R0(config-if)#ip add 192.168.10.65 255.255.255.224
R0(config-if)#no shut

R0(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

R0(config-if)#int g0/1
R0(config-if)#ip add 192.168.10.128 255.255.255.252
Bad mask /30 for address 192.168.10.128
R0(config-if)#ip add 192.168.10.129 255.255.255.252
R0(config-if)#no shut

R0(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

R0(config-if)#int g0/2
R0(config-if)#ip add 192.168.10.133 255.255.255.252
R0(config-if)#no shut

R0(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

R0(config-if)#int g0/3/0
R0(config-if)#ip add 192.168.10.137 255.255.255.252
R0(config-if)#no shut

%LINK-5-CHANGED: Interface GigabitEthernet0/3/0, changed state to down
R0(config-if)#

```

PC3:

IP Configuration	
☒ DHCP	☒ Static
IPv4 Address	192.168.10.94
Subnet Mask	255.255.255.224
Default Gateway	192.168.10.65
DNS Server	0.0.0.0

R1:

```

Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1
R1(config)#en secret 123
% Ambiguous command: "en secret 123"
R1(config)#enable secret 123
R1(config)#int g1/0
%Invalid interface type and number
R1(config)#int g0/1
R1(config-if)#ip add 192.168.10.130 255.255.255.252
R1(config-if)#no shut

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

R1(config-if)#int g0/0
R1(config-if)#ip add 192.168.10.1 255.255.255.192
R1(config-if)#no shut

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

R1(config-if)#

```

PC0:

IP Configuration	
☒ DHCP	☒ Static
IPv4 Address	192.168.10.62
Subnet Mask	255.255.255.192
Default Gateway	192.168.10.1
DNS Server	0.0.0.0

R2:

```

Router>
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R2
R2(config)#enab
R2(config)#enable se
R2(config)#enable secret 123
R2(config)#int g0/2
R2(config-if)#ip add 192.168.10.134 255.255.255.252
R2(config-if)#no shut

R2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

R2(config-if)#int g0/0
R2(config-if)#ip add 192.168.10.97 255.255.255.240
R2(config-if)#no shut

R2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

R2(config-if)#|

```

PC1:

IP Configuration	
☒ DHCP	☒ Static
IPv4 Address	192.168.10.110
Subnet Mask	255.255.255.240
Default Gateway	192.168.10.97
DNS Server	0.0.0.0

R3:

```

Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R3
R3(config)#enable secret 123
R3(config)#int g0/3/0
R3(config-if)#ip add 192.168.10.138 255.255.255.252
R3(config-if)#no shut

R3(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/3/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/3/0, changed state to up

R3(config-if)#int g0/0
R3(config-if)#ip add 192.168.10.113 255.255.255.240
R3(config-if)#no shut

R3(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

R3(config-if)#|

```

PC2:

IP Configuration	
☒ DHCP	☒ Static
IPv4 Address	192.168.10.126
Subnet Mask	255.255.255.240
Default Gateway	192.168.10.113
DNS Server	0.0.0.0

Konfiguracja OSPF:

R0:

```
R0>en
Password:
Password:
R0#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R0(config)#router ospf 1
R0(config-router)#network 192.168.10.64 0.0.0.31 area 0
R0(config-router)#network 192.168.10.128 0.0.0.3 area 0
R0(config-router)#network 192.168.10.132 0.0.0.3 area 0
R0(config-router)#network 192.168.10.136 0.0.0.3 area 0
R0(config-router)#
R0(config-router)#passive-interface g0/0
```

R1:

```
R1>
R1>en
Password:
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#ip ospf 1
^
% Invalid input detected at '^' marker.

R1(config)#router ospf 1
R1(config-router)#network 192.168.10.0 0.0.0.63 area 0
R1(config-router)#passi
R1(config-router)#passive-interface g0/0
R1(config-router)#do write
Building configuration...
[OK]
R1(config-router)#

R1(config-router)#network 192.168.10.128 0.0.0.3 area 0
R1(config-router)#do write
```

R2:

```

R2>en
Password:
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#router ospf 1
R2(config-router)#network 192.168.10.132 0.0.0.3 area 0
R2(config-router)#network 192.168.10.
00:28:36: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.10.137 on GigabitEthernet0/2 from
LOADING to FULL, Loading Done
96 0.0.0.15 area 0
R2(config-router)#pass
R2(config-router)#passive-interface a0/0

```

R3:

```

R3>en
Password:
R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#router ospf 1
R3(config-router)#network 192.168.10.136 0.0.0.3 area 0
R3(config-router)#network 192.168.10.112 0.0.0.15 area 0
R3(config-router)#passi
R3(config-router)#passive-interface
00:30:14: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.10.137 on GigabitEthernet0/3/0 from
LOADING to F
R3(config-router)#pas
R3(config-router)#passive-interface g0/0
R3(config-router)#

```

Testowanie:

PC3:

```

C:\>ping 192.168.10.62

Pinging 192.168.10.62 with 32 bytes of data:

Reply from 192.168.10.62: bytes=32 time<1ms TTL=126
Reply from 192.168.10.62: bytes=32 time=12ms TTL=126
Reply from 192.168.10.62: bytes=32 time=1ms TTL=126
Reply from 192.168.10.62: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.10.62:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 12ms, Average = 3ms

C:\>|

```

PC0:

```
C:\>ping 192.168.10.110

Pinging 192.168.10.110 with 32 bytes of data:

Reply from 192.168.10.110: bytes=32 time=1ms TTL=125
Reply from 192.168.10.110: bytes=32 time<1ms TTL=125
Reply from 192.168.10.110: bytes=32 time=24ms TTL=125
Reply from 192.168.10.110: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.10.110:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 24ms, Average = 6ms

C:\>ping 192.168.10.127

Pinging 192.168.10.127 with 32 bytes of data:

Reply from 192.168.10.138: bytes=32 time<1ms TTL=253
Reply from 192.168.10.138: bytes=32 time<1ms TTL=253
Reply from 192.168.10.138: bytes=32 time<1ms TTL=253
Reply from 192.168.10.138: bytes=32 time<1ms TTL=253

Ping statistics for 192.168.10.127:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

wszystko smiga :D